Code No. 3416

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CSE) IV-Semester (MAIN) Examinations Jun/Jul-2026

AUTOMATA LANGUAGES AND COMPUTATION

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	What is the precise role of the epsilon-closure of a state in an NFA with epsilon-transitions?	2	2	3
b)	Provide a simple example of a Context-Free Language (CFL) that cannot be accepted by any DPDA.	2	4	2
c)	Mention two decision algorithms applicable to CFGs that are decidable.	2	1	2
d)	What is a Multi-tape Turing Machine? How does its tape head setup work?	2	2	1
e)	State the undecidability status of determining if a context-free grammar is ambiguous.	2	1	4
f)	Predict what happens if you try to build an LR(0) parsing table for an ambiguous grammar.	2	4	2
2. a)	Consider the NFA with epsilon-transitions parameterized by the following state table:	6	3	3

δ	ϵ	a	b
$\rightarrow q_0$	$\{q_2\}$	$\{q_1\}$	$\{q_0\}$
q_1	$\{q_2\}$	$\{q_2\}$	$\{q_3\}$
q_2	$\{q_0\}$	$\emptyset$	$\emptyset$
q_3	$\emptyset$	$\emptyset$	$\{q_2\}$

Compute the complete E-closure for every state and convert this automation into an equivalent structural DFA using the subset construction method.

b)	Prove that if a DFA with n states accepts any string at all, it must accept a string of length less than n. Discuss how this fact supports the emptiness-testing algorithm.	6	5	2

(P.T.O.)

4. a)	Design a Non-Deterministic Pushdown Automaton (NPDA) that accepts the language of even-length palindromes: $L = \{ww^R \mid w \in \{a, b\}^*\}$ Provide the complete structural transition tables, specify your acceptance criteria, and trace the systematic computation allocations for the string w = abba.	6	6	1
b)	Construct a CFG that generates all valid strings over {a, b} containing an unequal number of a's and b's. Explain the structural logic of your production rules.	6	6	3
a)	Explain the concept of Greibach Normal Form (GNF). Convert the following grammar into GNF using systematic variable elimination: $S \rightarrow AA \mid a$ $A \rightarrow SS \mid b$	6	3	3
b)	Show how any context-free grammar can be transformed into an equivalent grammar without epsilon-productions, ensuring that L(G) -{epsilon} remains identical.	6	4	3
5. a)	Prove that a Multi-tape Turing Machine is computationally equivalent to a standard Single-tape Turing Machine. Describe the specific simulation algorithm that uses multiple tracks on a single tape to mimic the behavior of multiple independent tapes.	6	5	1
b)	Design a Turing Machine that computes the subtraction function f(m, n) = m - n for m ≥ n, and outputs 0 if m < n, using unary representations (1^m 0 1^n).	6	6	1
6. a)	Explain the Post Correspondence Problem (PCP). Determine if the following instances of PCP have a matching solution sequence: List A = (10, 011, 101) List B = (101, 11, 011) Show your step-by-step structural index selection trace.	6	4	4
b)	Compare regular grammars, context-free grammars, context-sensitive grammars, and unrestricted grammars.	6	4	2
7. a)	Explain the acceptance of languages by PDA using final states and empty stack methods.	6	2	1
b)	Compare DFA, NFA, Moore Machine, and Mealy Machine with suitable examples.	6	4	1

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